

MATH 195: 9/24
INVERSES OF FUNCTIONS

Combining functions. When talking about functions, it can be helpful to distinguish the function f from a formula $f(x)$ for its output. While it's common to write $f(x)$ for both, sometimes the extra parentheses everywhere makes it harder to see what's going on.

We can combine numbers with arithmetic operations. The same operations can be used to combine functions:

$$f + g, \quad f - g, \quad f \cdot g, \quad f/g.$$

With functions we have a new way of combining that doesn't make sense with numbers. The *composition* $f \circ g$ of two functions f and g is what you get if you first do one function and use its output as the input to the other function:

$$(f \circ g)(x) = f(g(x)).$$

You get a formula for $f(g(x))$ by substituting the formula for $g(x)$ as the input u to $f(u)$.

Geometric transformations.

These are examples of composition. If $f(x)$ is a function and $\ell(x)$ is a linear function (with nonzero slope), then $f(\ell(x))$ is a horizontal transformation of $f(x)$ and $\ell(f(x))$ is a vertical transformation of $f(x)$.

- (1) Consider $\ell(x) = -x + 2$. What transformations are applied to $f(x) = \frac{1}{x}$ to get the function $\ell(f(x))$?
[Hint: write out a formula for $\ell(f(x))$.]
- (2) Consider $k(x) = x + 4$. What transformation is applied to $g(x) = x^2$ to get the function $g(k(x))$?
- (3) Consider $\ell(x)$ and $k(x)$ from the previous two problems. What transformations are applied to $h(x) = x^3$ to get the function $\ell(h(k(x)))$?

The algebra of composition. Composition is *associative*:

$$(f \circ g) \circ h = f \circ (g \circ h).$$

Think like multiplication: if you multiply three numbers it doesn't matter which two you combine first, the final product is the same.

But composition is not *commutative*:

$$f \circ g \neq g \circ f, \quad \text{for most } f \text{ and } g.$$

Think like subtraction: the order of subtraction matters for the outcome.

- (1) Check an instance of associativity, for the functions

$$f(x) = 2x - 1, \quad g(x) = \frac{1}{x}, \quad h(x) = x^2.$$

That is, compute formulas for $(f \circ g) \circ h$ and $f \circ (g \circ h)$ and see that they are equivalent.

- (2) Check an instance of the failure of commutativity, for the functions

$$f(x) = \frac{1}{x-1}, \quad g(x) = x^2.$$

That is, compute formulas for $f \circ g$ and $g \circ f$ and explain why they are not equivalent.

- (3) Confirm that sometimes commutativity holds by looking at the functions

$$f(x) = x^3, \quad g(x) = x^{-2}.$$

That is, check that $f \circ g$ and $g \circ f$ are the same in this case.

- (4) Write down random formulas for functions f and g . Do you get $f \circ g = g \circ f$?
- (5) For subtraction you know order matters. Are there any cases where the order doesn't matter? That is, can you find any numbers x and y so that $x - y = y - x$? What about the analogous question for division?

Identities and inverses. Adding and multiplying have associated concepts of opposites. The *additive inverse* of a number x is its negative $-x$ and the *multiplicative inverse* of a number x is its reciprocal $\frac{1}{x}$. These concepts also make sense for functions. The additive inverse of f is $-f$ and the multiplicative inverse of f is $1/f$. There's two main ways to think about inverses.

- An inverse is the backwards or opposite version. The opposite of adding x^2 is subtracting x^2 , so the additive inverse of x^2 is $-x^2$. The opposite of multiplying by 7 is dividing by 7, so the additive inverse of 7 is $\frac{1}{7}$.
- Combining something with its inverse gives you the *identity*, the “do-nothing” object. For addition, the identity is 0; adding by 0 doesn't change the value of a function. So $-x^2$ is the additive inverse of x^2 because $x^2 + (-x^2) = 0$. The *multiplicative identity* is 1. And $\frac{1}{7}$ is the multiplicative inverse of 7 because $7 \cdot \frac{1}{7} = 1$.

We think of inverses with regard to composition—we just say *inverses*—in both of these ways. Here's the first.

- The inverse of a function f is the function f^{-1} you get by running the function backward. That is, you swap the role of the inputs and outputs.
- Thinking in terms of visualizations of functions, you reverse the direction of the arrows.
- In formulas, $f(x) = y$ is equivalent to the equation $f^{-1}(y) = x$. To find a formula for $f^{-1}(y)$, solve the equation $f(x) = y$ for x in terms of y .
- In terms of graphs, you swap the x - and y -axes. This is equivalent to saying you reflect the graph across the diagonal line $y = x$.

Here is the second.

- The *identity* function, commonly written id , is the “do-nothing” function which outputs the input with no change. That is, $\text{id}(x) = x$.
- Two functions f and f^{-1} are inverses of each other if

$$f \circ f^{-1} = f^{-1} \circ f = \text{id}.$$

Think: if you run a function, then do the opposite, you get back the input you started with.

When do inverse functions exist? Inverses don't always exist. Think of how 0 doesn't have a multiplicative inverse. With functions and composition, it is very common for inverses not to exist.

The problem is, if two different inputs x_1 and x_2 go to the same output y under a function f , then how do we reverse it. Given y an input for f^{-1} , should the output should be x_1 or x_2 ? We'd have to have two different outputs for one input, but then f^{-1} wouldn't be a function. It wouldn't pass the vertical line test.

It turns out this is the only obstacle.

- Call a function f *one-to-one* if different outputs go to different inputs.

$$x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$$

- Equivalently, a function is one-to-one if whenever two outputs are the same they had to have come from the same input.

$$f(x_1) = f(x_2) \implies x_1 = x_2$$

This gives us an algebraic process to check whether f is one-to-one. Start with the equation $f(x_1) = f(x_2)$. If you can turn that into the equation $x_1 = x_2$, then it is one-to-one.

- Graphically, a function is one-to-one if it passes the *horizontal line test*. Every horizontal line intersects the graph in at most one place.
- If a function is increasing everywhere or decreasing everywhere then it is one-to-one.

Key Fact: A function has an inverse if and only if it is one-to-one.

- (1) Compute a formula for the inverse of $f(x) = 3x - 2$. Check that $f(f^{-1}(x)) = x$.
- (2) Compute a formula for the inverse of $g(x) = \frac{3}{2x-1} + 2$. Check that $g(g^{-1}(x)) = x$.
- (3) Explain why $h(x) = 4(x-2)^2 - 1$ isn't one-to-one by looking at the graph.
- (4) Algebraically explain why $h(x)$ isn't one-to-one. That is, start with the equation $h(x_1) = h(x_2)$ and show why the equation can't be reduced to $x_1 = x_2$.
- (5) What happens if you try to find a formula for $h^{-1}(x)$ by solving the equation $h(x) = y$ for x ? Explain why what you get shows you that h doesn't have an inverse.
- (6) Give an example, either as a graph or a formula, of a function which isn't one-to-one because infinitely many inputs go to the same output.
- (7) Explain why $f \circ \text{id} = \text{id} \circ f = f$ for any function f .